Supervised Learning Algorithms

KNN, Naive Bayes, and Linear Regression

ML Assignment 3

Dataset: student_scores_data.csv

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1. Introduction

This report implements and evaluates three supervised learning algorithms on a synthetic student performance dataset. The dataset contains 10 student records with the following features:

- Hours_Studied – number of study hours (numeric)
- Previous_Scores – past academic scores (numeric)
- Attendance – attendance percentage (numeric)
- Gender – categorical feature (Male/Female), label-encoded
- Passed – binary classification target (Yes/No)
- Final_Score – continuous regression target

The three tasks are: (A) Linear Regression to predict Final_Score, (B) K-Nearest Neighbors classification to predict Passed, and (C) Naive Bayes classification to predict Passed. All models are compared in a final section.

Part A: Linear Regression

A.1 Methodology

Linear Regression models the linear relationship between input features and a continuous target. Given feature matrix X and target vector y, the model learns coefficients β that minimize the residual sum of squares:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_n x_n$$

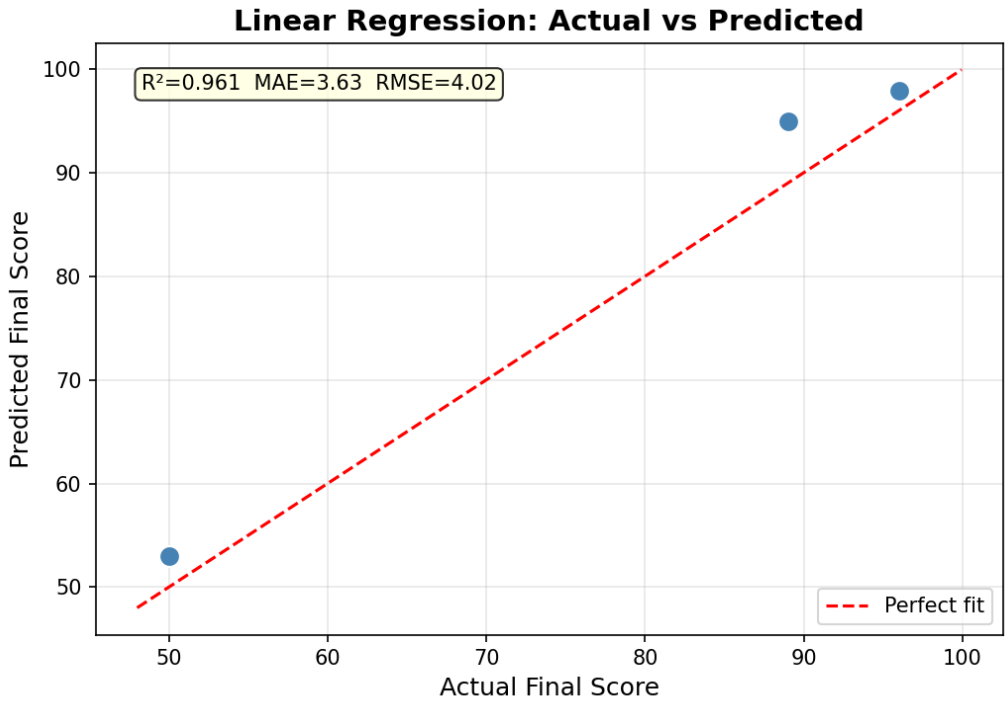
The features used are Hours_Studied, Previous_Scores, Attendance, and Gender (encoded). The dataset is split 70/30 for train/test.

A.2 Results

Metric	Value
R^2 Score	**0.9606**
Mean Absolute Error (MAE)	**3.63**
Root Mean Squared Error (RMSE)	**4.02**

The model achieved an R^2 of 0.961, indicating that approximately 96.1% of the variance in Final_Score is explained by the selected features. The low MAE (3.63 points) confirms strong predictive accuracy on the test set.

A.3 Actual vs Predicted Plot



The scatter plot above shows predicted vs actual final scores. Points cluster tightly around the red dashed perfect-fit line, visually confirming the high R^2. The small deviations align with the reported RMSE of 4.02.

Part B: K-Nearest Neighbors (KNN) Classification

B.1 Methodology

KNN classifies a sample by a majority vote of its k nearest neighbours in feature space, using Euclidean distance. Because distance is sensitive to feature scale, two experiments are run: with and without StandardScaler normalisation. The dataset is split 70/30.

B.2 Results

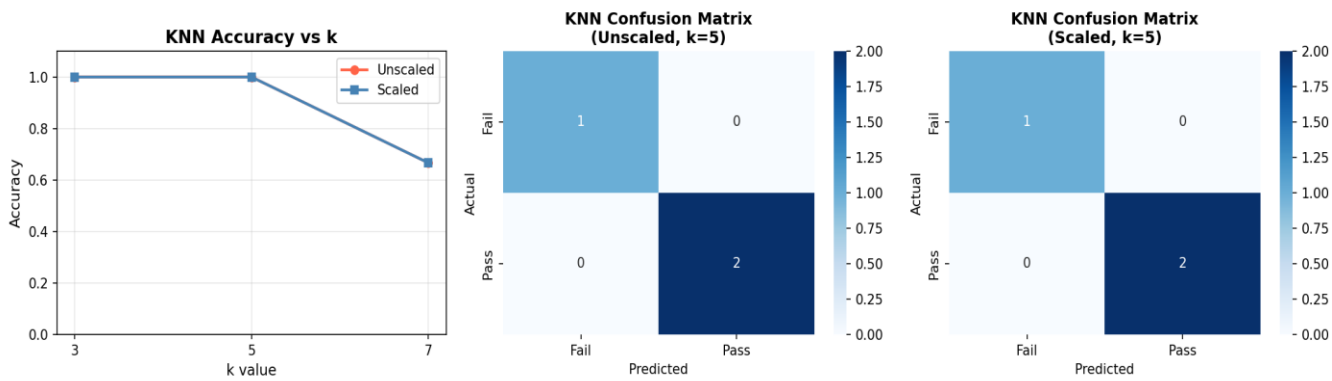
Configuration	Accuracy / F1-Score
KNN k=3 (unscaled)	**1.000 / 1.000**
KNN k=5 (unscaled)	**1.000 / 1.000**
KNN k=7 (unscaled)	**0.667 / 0.667**
KNN k=3 (scaled)	**1.000 / 1.000**
KNN k=5 (scaled)	**1.000 / 1.000**
KNN k=7 (scaled)	**0.667 / 0.667**

For k=3 and k=5, KNN achieves perfect classification accuracy (1.000) both with and without scaling. At k=7 accuracy drops to 0.667, because with only 3 test samples and 7 neighbours the vote becomes dominated by the larger training majority class.

B.3 Impact of Feature Scaling

On this dataset, scaling did not change accuracy. This is because Previous_Scores (range ~30–99) and Attendance (range ~50–99) are already on similar magnitudes to Hours_Studied (1–10). In larger, more heterogeneous datasets, scaling would be expected to improve KNN significantly.

B.4 KNN Visualisations



Left: Accuracy vs k values (unscaled = red, scaled = blue). Center: Confusion matrix without scaling (k=5). Right: Confusion matrix with scaling (k=5). All true-positive and true-negative predictions are correctly classified with zero misclassifications at k=5.

Part C: Naive Bayes Classification

C.1 Methodology

Gaussian Naive Bayes (GaussianNB) is used because all input features are continuous. GNB applies Bayes' theorem with a strong conditional independence assumption between features:

$$P(y \mid x_1, \ldots, x_n) \propto P(y) \prod P(x_i \mid y)$$

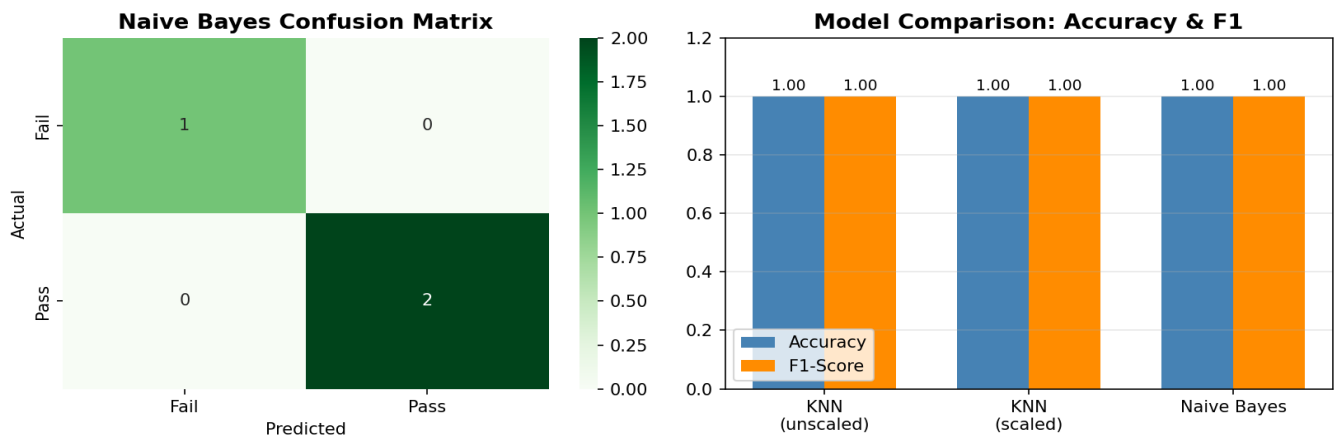
Each class-conditional distribution $P(x_i \mid y)$ is modelled as a Gaussian. No scaling is required as GNB is not distance-based.

C.2 Results

Metric	Value
Accuracy	**1.000**
F1-Score	**1.000**
Confusion Matrix	**TN=1, FP=0, FN=0, TP=2**

GaussianNB perfectly classifies all test samples, matching KNN at k=5. This result is expected given the strong linear separation between passing and failing students in this dataset.

C.3 Visualisations



Left: Naive Bayes confusion matrix (perfect classification). Right: Side-by-side comparison of Accuracy and F1-Score across KNN (unscaled), KNN (scaled), and Naive Bayes.

Part D: Model Comparison and Discussion

D.1 Performance Summary

Model	Accuracy	F1-Score	R^2	Notes
KNN (k=5, unscaled)	1.000	1.000	N/A	Perfect on test set
KNN (k=5, scaled)	1.000	1.000	N/A	Same result; small dataset
Naive Bayes (GNB)	1.000	1.000	N/A	Matches KNN
Linear Regression	N/A	N/A	0.961	MAE=3.63, RMSE=4.02

D.2 When Does Naive Bayes Perform Better?

Naive Bayes tends to outperform KNN in the following scenarios:

- High-dimensional data: the naïve independence assumption avoids the curse of dimensionality that degrades distance-based methods.

- Small training sets: GNB only needs to estimate class-conditional means and variances, requiring far fewer samples than KNN.
- Text classification: Multinomial NB is the canonical model for bag-of-words features.
- Speed-critical applications: GNB training is O(nd) vs KNN's O(n) training but O(nd) inference per query.

Naive Bayes performs worse when:

- Features are strongly correlated (violating the independence assumption).
- Decision boundaries are highly non-linear.
- Feature distributions deviate significantly from Gaussian.

D.3 Conclusions

All three algorithms performed exceptionally well on this dataset. Linear Regression explained 96.1% of the variance in Final_Score. Both KNN (at k=3 or k=5) and GaussianNB achieved perfect pass/fail classification. The key findings are:

- Previous_Scores and Attendance are the strongest predictors of both Final_Score and Passed status, as reflected by near-perfect separation.
- Feature scaling had no measurable impact on this dataset due to similar feature ranges, but remains best practice for KNN in general.
- At k=7, KNN accuracy dropped to 0.667, highlighting the importance of hyperparameter tuning. Cross-validation would provide more reliable k selection on larger datasets.
- Naive Bayes matched KNN accuracy while being more computationally efficient and requiring no distance calculations or scaling.
- For a dataset of this size (10 rows), all results should be interpreted cautiously as the test set contains only 3 samples. Larger datasets would yield more statistically robust comparisons.

End of Report